Regulating AI Security in New Zealand's Energy Sector:
A Proposal for the AI Critical Energy Infrastructure Security Regulation (2026)

Introduction

AI is becoming more integrated into modern energy systems. AI technologies are used in New Zealand to balance grid loads, forecast electricity demand, optimize renewable energy, perform predictive maintenance, and detect faults automatically. As the country moves toward more renewable generation and independent energy production, digital grid management systems play an important role in ensuring reliability and efficiency. While AI improves performance and flexibility, it also raises new security concerns. Energy systems are considered critical infrastructure because disruptions endanger public safety, economic stability, and national security (International Energy Agency [IEA], 2021).
"Country selection and topic analysis were developed with AI assistance" (Kimi AI interaction, February 7, 2026; see Appendix A).

In New Zealand, electricity powers hospitals, emergency services, telecommunications networks, financial markets, transportation systems, and military operations. A large-scale grid failure could have serious societal consequences. New Zealand has taken a flexible, innovative approach to AI governance. Rather than implementing comprehensive AI legislation, the government relies on existing regulatory frameworks and international principles (Ministry of Business, Innovation and Employment [MBIE], 2025).

While this approach promotes technological growth, it leaves gaps in high-risk areas where AI failures could have severe consequences. AI systems have several vulnerabilities, including adversarial attacks, data poisoning, model manipulation, and automated decision-making errors (National Cyber Security Centre [NCSC], 2023). These risks are distinct from those faced by traditional software systems because machine learning models rely on training data and adaptive processes that can be difficult to audit or fully understand. Researchers have found that AI in critical infrastructure increases systemic risk due to model opacity, automation speed, and reliance on large-scale data integration (Brundage et al., 2018).

Given the serious risks, this paper proposes a targeted regulatory response: **The Artificial Intelligence Critical Energy Infrastructure Security Regulation (2026).** This proposal focuses on high-risk AI systems used in New Zealand's operational energy infrastructure. The regulation aims to improve defense and accountability while remaining fair and encouraging innovation.

New Zealand's Regulatory Context

Energy Sector Governance

New Zealand has a parliamentary system in which legislation is passed by Parliament and implemented by ministries and independent regulatory agencies. Several institutions hold regulatory authority in the energy sector. The **Electricity Authority** regulates the electricity market to ensure competition, reliability, and efficiency. The **Ministry of Business, Innovation and Employment (MBIE)** oversees and develops national energy policy. **Transpower,** a state-owned enterprise, manages the national transmission grid and ensures system stability. The **Commerce Commission** regulates infrastructure networks for economic purposes.

The **National Cyber Security Centre (NCSC)** maintains cybersecurity oversight, providing critical infrastructure operators with threat intelligence and incident response support. Although these agencies regulate electricity markets and general cybersecurity standards, no current legislation implements AI-specific requirements for grid control systems. The Electricity Industry Act 2010 covers market structure and operational reliability but does not address machine learning vulnerabilities (New Zealand Legislation, 2010).

AI Governance in New Zealand

New Zealand aligns its AI governance strategy with the OECD AI Principles, which emphasize robustness, transparency, safety, and accountability (Organization for Economic Co-operation and Development [OECD], 2019). Government strategy documents indicate that regulation should be introduced when risks are "acute" and not adequately mitigated by existing frameworks (MBIE, 2025).

AI deployment in energy infrastructure meets this requirement. Because electricity is essential to almost every service, disruptions could have national consequences. While cybersecurity laws apply broadly, they do not specifically address adversarial machine learning or AI system integrity. A dedicated regulation would bring clarity and establish consistent standards throughout the industry.

Security Risks of AI in Energy Infrastructure

"Technical definitions and examples of adversarial attacks and data poisoning were clarified using AI-assisted research." (Kimi, February 8, 2026; see Appendix A).

AI-Specific Cyber Vulnerabilities

Digitization has already increased cybersecurity risks in the energy industry (IEA, 2021). AI systems introduce vulnerabilities that are not present in traditional software systems. **Data poisoning** attacks can manipulate training datasets, causing AI systems to generate incorrect results. Adversarial examples, or subtle manipulations of input data, can lead to machine learning models misclassifying or misinterpreting signals (Goodfellow, Shlens, & Szegedy, 2015). In energy grids, such manipulation may disrupt load

balancing or fault detection. When AI systems manage operational infrastructure, these flaws can have physical consequences.

Traditional cybersecurity strategies frequently focus on network intrusion prevention. However, AI security requires additional protections related to model integrity, data provenance, and algorithmic transparency (Brundage et al., 2018). The current regulatory framework does not explicitly require these safeguards in energy infrastructure.

Recurrent Operational Failures

Electricity grids are complicated, interconnected systems. AI systems used for real-time grid balancing run at high speeds. If an AI model produces incorrect results, whether due to a flaw, insufficient training data, or adversarial interference, the resulting imbalance can spread quickly throughout the network. Energy system failures are rarely isolated incidents. Small disruptions can spread across transmission lines and power plants. According to the IEA (2021), automation increases both efficiency and systemic interdependence, potentially amplifying the impact of errors. Without safety measures, AI systems may act too quickly for humans to detect mistakes. Because some machine learning models operate as closed systems, determining the root cause of failure can be difficult. This lack of transparency emphasizes the importance of pre-deployment auditing and ongoing monitoring.

Supply Chain and Vendor Risks

AI systems in energy infrastructure frequently rely on third-party vendors, cloud providers, and globally developed models. These dependencies create vulnerabilities in the supply chain. Without transparency requirements, regulators may not be able to see software dependencies or AI components developed by foreign companies. Researchers emphasize the importance of AI supply chain integrity in national security, particularly in critical infrastructure contexts (Brundage et al., 2018). Given New Zealand's reliance on imported technologies, oversight mechanisms are required to manage geopolitical risk.

National Security Implications

Electricity is essential for healthcare systems, emergency response, financial markets, and military capabilities. The IEA (2021) identifies energy cybersecurity as a strategic priority for modern states. AI vulnerabilities are an evolving threat channel that adversarial actors may exploit. Because AI-driven systems can adapt autonomously, detecting and mitigating failures is more difficult than in traditional software systems. Therefore, binding standards are required to ensure consistent protection across operators.

Proposed Regulation: AI Critical Energy Infrastructure Security Regulation (2026)

"The initial structure of the proposed regulatory components was refined using AI-assisted drafting" (Kimi, February 8, 2026; see Appendix)**.**

Purpose:

The **AI Critical Energy Infrastructure Security Regulation (2026)** is designed to address the risks associated with AI systems used in energy infrastructure. The regulation aims to:

1. Protect New Zealand's electricity system from AI-related cybersecurity threats.
2. Increase operational resilience in high-risk infrastructures.
3. Develop clear accountability mechanisms for AI deployment.
4. Improve transparency in AI supply chains.
5. Maintain public trust in important national systems.
6. Ensure Māori data sovereignty and consultation rights when AI impacts tribal lands or resources.

Unlike broad AI legislation, this regulation is sector-specific and risk-driven. It does not govern all AI applications. Instead, it targets AI systems whose failure, compromise, or manipulation could cause significant disruptions to the national electricity supply. This proportional approach reflects the government's stated principle that regulation should be implemented when risks are severe and cannot be adequately mitigated by existing frameworks (MBIE, 2025).

Scope and Applicability

The regulation applies to:

- Companies involved in electricity generation.
- Transmission operators (including Transpower, a regulated entity).
- Distribution network operators.
- Third-party vendors provide AI systems for operational infrastructure.

The regulation includes AI systems used for:

- Grid load balancing.
- Demand forecasting.
- Automated fault detection.
- Real-time monitoring of infrastructure.
- Operational control decisions that affect generation or transmission.

Exemptions:

- AI systems are used only for administrative, financial, or marketing purposes.
- AI should be tested in a safe environment rather than in live electrical systems.
- Microgrids with less than 10 MW capacity.

Definitions

AI system: A computational system that uses machine learning, predictive modeling, or automated decision-making to improve energy infrastructure operations.

High-Risk AI System: An AI system in charge of grid balancing, fault response, or emergency operations in which failure could affect more than 50,000 customers or national security.

Adversarial Attack: Malicious input designed to manipulate AI models into producing incorrect results.

Iwi Authority: A recognized Māori tribal governance body with authority over traditional lands and resources.

Māori Data Sovereignty: Māori tribes have the right to collect, use, and manage data about their people, lands, and resources, including geothermal and energy data.

Critical Energy Infrastructure: Facilities and digital systems are critical to electricity generation, transmission, and distribution in New Zealand.

Core Requirements

Mandatory AI Security Risk Assessments

Before deployment, operators must conduct structured AI security risk assessments on all high-risk systems. These assessments must evaluate the following:

- Vulnerability to adversarial attacks.
- Data integrity safeguards.
- Model robustness in stressful conditions.
- Possible recurrent failure impacts.
- Human oversight mechanisms.

Risk assessments must be submitted to the Electricity Authority before operations begin. Annual reassessments ensures that evolving grid conditions and updated threat intelligence are considered.

Independent Security Certification

High-risk AI systems must be subjected to independent third-party cybersecurity audits before deployment. Certification should confirm:

- Resistance to adversarial input.
- Secure model training pipelines.
- Controlled access permissions.
- Redundancy protocols that fail safely.
- Human override functionality has been verified.

The NCSC Guidelines for Secure AI System Development (NCSC, 2023) would transition from advisory guidance to enforceable compliance standards under this regulation.

Human Oversight Requirements

All emergency decisions must have human approval where AI cannot act on its own during a crisis. This requirement maintains human accountability for critical decisions affecting national infrastructure, rather than eliminating automation overall. In emergency situations, human oversight reduces automation bias and prevents rapid progressive errors.

Continuous Monitoring and Anomaly Detection

Operators must implement real-time monitoring systems that can detect:

- Abnormal AI output behavior.
- Unexpected input anomalies.
- Indicators of adversarial interference. ● Model performance degradation.

When the AI system sends an alert, humans must follow specific written instructions to check and respond.

Incident Reporting Requirement

AI-related cybersecurity incidents must be reported within 72 hours to the National Cyber Security Centre. Incidents that can be reported include:

- Confirmed cyber breaches.
- Significant AI malfunction.
- A couple of recurring system failures.
- Detecting adversarial manipulation attempts.

Supply Chain Transparency

Operators must disclose:

- The origins of AI models.
- Vendor relationships.
- Foreign dependencies.
- Critical software components.

Māori Consultation and Data Sovereignty

- When deploying AI systems on Māori land or using tribal resources, operators must inform affected iwi authorities within 72 hours.
- Geothermal data on Māori-owned land requires explicit data governance agreements that respect tikanga Māori.

Enforcement and Oversight

Issuing Body: Joint authority of the **Electricity Authority (**operational oversight**)** and **National Cyber Security Centre** (cybersecurity standards). This creates new coordinated enforcement, as the NCSC currently lacks independent regulatory authority.

Enforcement tools:

- Annual Compliance Reporting.
- Random security audits.
- Mandatory certification reviews.
- On-site inspections.

Penalties for non-compliance:

- Civil fines of up to **NZD $5 million** for systematic violations.
- Suspension of AI system operations.
- Mandatory shutdown orders for serious safety threats.
- Criminal charges for reckless endangerment resulting in actual risk of grid failure.
- Public disclosure of violations (unless the national security exemption applies).

Example Use Case

Scenario: Transpower uses AI to automate grid balancing during peak renewable generation.
Application:

- Pre-deployment risk assessment determines vulnerability to adversarial input manipulation.
- Experts test that a human can stop the AI and take over in just 30 seconds.

- Power lines near Māori-owned geothermal areas require approval from tribal leaders.
- Real-time monitoring detects anomalous load predictions and allows human review.

- When anomalous data suggest potential poisoning, operators override AI and report to the NCSC within 72 hours.
- Supply chain disclosure reveals foreign cloud dependency, which requires the installation of backup systems.

Outcome: The regulation promotes grid stability, protects Māori interests, and maintains national security.

Implementation Timeline

Phase	Date	Requirement
Implementation	July 1, 2026	Regulation takes effect, and reporting obligations begin.
Transition period	July 1, 2026 - November 1, 2027	Existing AI systems must achieve compliance, while new systems comply upon deployment.
First review	July 1, 2029	Mandatory regulation review.
Follow-up Reviews	Every 2 years	Continuous reassessment of standards and technological developments.

Addressing Counterarguments

1. **Regulation will slow innovation in renewable energy.**
   Response: The regulation only covers high-risk AI systems used in operational grid control, not research or early-stage development. Innovation continues, but deployment requires safety. The transition period allows existing systems to meet compliance requirements without disrupting operations.

2. **Energy providers will bear the burden of compliance costs.**
   Response: The financial costs of a major shutdown exceeds compliance costs. The IEA (2021) documents that global critical infrastructure cyberattacks cause billions in economic damage. Providing security lowers long-term risk and liability.

3. **Existing cybersecurity laws are sufficient.**
   Response: Traditional cybersecurity laws do not explicitly address adversarial machine learning model manipulation or algorithmic opacity (Goodfellow et al., 2015; NCSC, 2023). AI systems introduce qualitatively new vulnerabilities that require specific safeguards.

4. **Human oversight reduces efficiency.**
   Response: Human-controlled systems are only necessary for critical emergency decisions. Routine automated operations continue without interruption. The requirement ensures accountability in situations where errors could cause nationwide disruption.

Conclusion

AI has significant benefits for New Zealand's energy sector, particularly in improving efficiency, integrating renewable generation, and optimizing grid performance. However, incorporating AI into operational control systems involves new and evolving security risks. Machine learning vulnerabilities, adversarial attacks, model variations, and supply chain dependencies create risks that traditional cybersecurity laws do not fully address. Failure in regulating electricity infrastructure creates a significant risk to public safety, economic stability, and national security, especially given New Zealand's obligations to Māori data sovereignty. The AI Critical Energy Infrastructure Security Regulation (2026) focuses on high-risk AI systems in operational energy infrastructure.

The regulation promotes resilience without affecting innovation by requiring risk assessments, independent certification, human oversight, real-time monitoring, incident reporting, supply chain transparency, Māori consultation, and frequent regulatory reviews. It distinguishes between high-risk operational AI and lower-risk applications, allowing flexibility while ensuring safety. Strong AI security rules are necessary to protect New Zealand's future, maintain tech leadership, and honor Māori rights as energy systems become more digital.

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